Customer Shopping Behavior Analysis

Uncovering insights from 3,900 purchases to guide strategic business decisions.

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Dataset Overview

3,900

Transactions

Purchase records analyzed

18

Features

Data points per transaction

37

Missing Values

Review ratings imputed

Key features include customer demographics, purchase details, and shopping behavior metrics.

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Data Preparation Process

01

Data Loading

Imported dataset using pandas

02

Exploration

Checked structure and summary statistics

03

Cleaning

Handled missing values and standardized columns

04

Feature Engineering

Created age groups and purchase frequency metrics

05

Database Integration

Loaded cleaned data into PostgreSQL

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SQL Analysis: Revenue Insights

Gender Revenue Split

Male: $157,890

Female: $75,191

Male customers generated over double the revenue of female customers.

Age Group Performance

Young Adult: $62,143

Middle-aged: $59,197

Adult: $55,978

Senior: $55,763

Young adults lead in revenue contribution.

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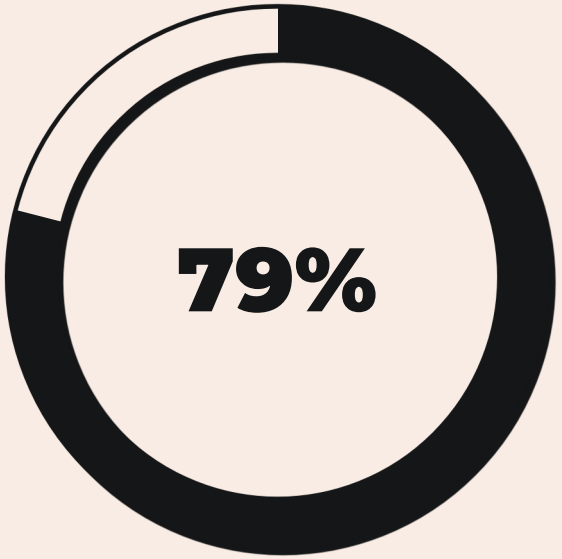
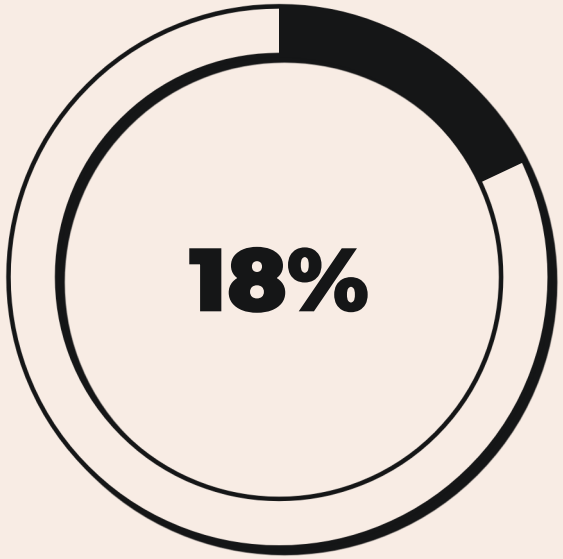
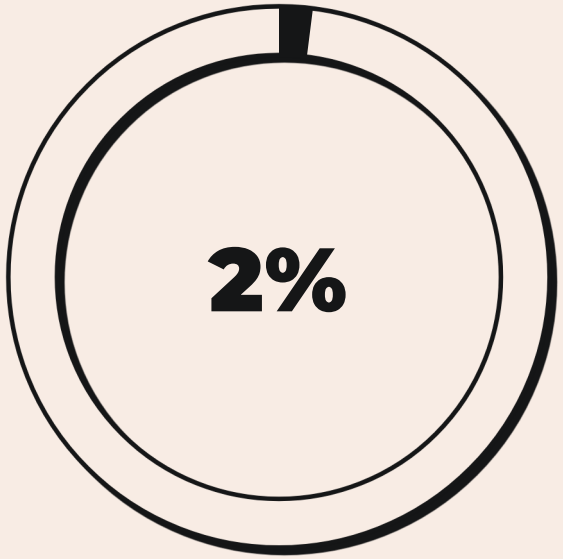
Customer Segmentation







Loyal Customers

3,116 customers with consistent purchase history

Returning

701 customers making repeat purchases

New

83 first-time buyers

Highly loyal customer base with strong repeat purchase behavior.

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Product Performance

Top-Rated Products

- Gloves: 3.86 rating
- Sandals: 3.84 rating
- Boots: 3.82 rating

Best-Selling Categories

- Clothing: 1,737 purchases
- Accessories: 504 purchases
- Footwear: 455 purchases

Most-Discounted Items

- Hat: 50% discounted
- Sneakers: 49.66% discounted
- Coat: 49.07% discounted

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Subscription Analysis

Subscriber Base

1,053 subscribers (27%) vs. 2,847 non-subscribers (73%)

Average Spend

Subscribers: $59.49 | Non-subscribers: $59.87

Total Revenue

Subscribers: $62,645 | Non-subscribers: $170,436

Shipping Comparison

Standard: $58.46 average

Express: $60.48 average

Repeat Buyer Pattern

958 repeat buyers subscribed, while 2,518 did not

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Power BI Dashboard

Key Metrics

3,900 customers, $59.76 average purchase, 3.75 average rating

Interactive Filters

Subscription status, gender, category, and shipping type

Visualizations

Donut charts, bar charts, and horizontal bar charts

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Key Findings

- Male customers drive 68% of total revenue
- Young adults generate highest revenue at $62,143
- 839 customers used discounts while spending above average
- Only 27% of customers are subscribed to promotions

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Strategic Recommendations

Boost Subscriptions

Promote exclusive benefits for subscribers

Customer Loyalty Programs

Reward repeat buyers to move them into "Loyal" segment

Review Discount Policy

Balance sales boosts with margin control

Product Positioning

Highlight top-rated and best-selling products in campaigns

Targeted Marketing

Focus on high-revenue age groups and express-shipping users

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